

Agentic Orchestration Engineer

Delivery Plan · 8 Sessions × 2.5 Hours

3.A Agentic Systems Integration › 3.B Agentic Systems Business Analysis › Capstone & Shark Tank

Track 3 is for participants who will design and integrate multi-agent systems with enterprise stacks. 3.A covers the engineering — single-agent and multi-agent orchestration, MCP server authoring, enterprise SaaS integration and observability. 3.B covers business analysis for agentic systems — discovery, requirements, business case and engineering handoff.

01 SPECIALIZATION HIGHLIGHTS

Specialization	Agentic Orchestration Engineer
Focus	Designing and integrating multi-agent systems with enterprise stacks.
Structure	Two parts — 3.A Agentic Systems Integration + 3.B Agentic Systems Business Analysis
Duration	20 hours · 8 sessions × 2.5 hours · delivered over 8 days
Cadence	3.A across 4 sessions, then 3.B across 4 sessions; a hands-on lab in every session
Delivery	Instructor-led, with applied hands-on in each session
Assessment	1-hour assessment after each part
Capstone	Group project (max 3) · 10–15 days · defended Shark-Tank style
Capstone domains	Insurance · Banking · Healthcare · Supply Chain · Logistics

02 DELIVERY PLAN — PART 3.A

3.A Agentic Systems Integration

10 HOURS · 4 SESSIONS

Single- and multi-agent orchestration, MCP authoring, enterprise integration and observability.

Every session runs the same rhythm — concept framing, an instructor-led walkthrough, a guided hands-on lab, and a short review — within 2.5 hours.

SESSION 1 Day 1

2.5 hours

Objective Master single-agent engineering with LangGraph and compare vendor-native SDKs.

Topics Covered	Hands-On Lab (Applied)
▪ Single-agent mastery – LangGraph with LangGraph Studio, PostgreSQL checkpointing, HITL nodes, typed tools with Pydantic AI ▪ Vendor-native alternatives – Claude Agent SDK, OpenAI Agents SDK – when to pick each	▪ Build a stateful single-agent with LangGraph + PostgreSQL checkpointing, a HITL gate, typed tools and full LangSmith trace observability ▪ Rebuild the same agent with Claude Agent SDK; compare ergonomics, MCP integration and trace quality
SESSION FLOW Concept framing & instructor walkthrough ~60 min · Guided hands-on lab ~75 min · Review, Q&A & wrap-up ~15 min	

SESSION 2 Day 2

2.5 hours

Objective Orchestrate multi-agent systems and author custom MCP servers.

Topics Covered	Hands-On Lab (Applied)
▪ Multi-agent orchestration – CrewAI for role-based prototyping, AG2 for production multi-agent, Microsoft Agent Framework ▪ MCP – authoring custom MCP servers with auth and rate limiting	▪ Design and implement a 3-agent orchestration (planner, researcher, validator) using LangGraph multi-agent with shared memory and typed contracts ▪ Rebuild the multi-agent flow in CrewAI; discuss when to pick which framework
SESSION FLOW Concept framing & instructor walkthrough ~60 min · Guided hands-on lab ~75 min · Review, Q&A & wrap-up ~15 min	

SESSION 3 Day 3

2.5 hours

Objective Apply A2A and integrate agents with enterprise SaaS systems.

Topics Covered	Hands-On Lab (Applied)
▪ A2A (Agent-to-Agent) Protocol – cross-vendor multi-agent orchestration ▪ Enterprise SaaS integration – Salesforce, ServiceNow, SAP Ariba via MCP and MuleSoft CloudHub / Zapier Central	▪ Author a custom MCP server exposing a domain knowledge base with auth, rate limiting and schema discipline; consume it from the agent ▪ Integrate the agent with mock Salesforce and ServiceNow via MuleSoft CloudHub; verify OAuth2 flow and circuit-breaker behaviour
SESSION FLOW Concept framing & instructor walkthrough ~60 min · Guided hands-on lab ~75 min · Review, Q&A & wrap-up ~15 min	

SESSION 4 Day 4**2.5 hours****Objective** Add agent observability and engineer resilience against partial failures.

Topics Covered	Hands-On Lab (Applied)
▪ Observability for agents – LangSmith tracing at scale, audit logs, token-budget enforcement, prompt-injection guards ▪ Resilience and partial-failure handling – retry, idempotency, circuit breakers, escalation routing, HITL, failure recovery	▪ Add retry, idempotency, escalation routing and token-budget guards; run a chaos drill simulating a partial enterprise SaaS outage
SESSION FLOW Concept framing & instructor walkthrough ~60 min · Guided hands-on lab ~75 min · Review, Q&A & wrap-up ~15 min	

03 DELIVERY PLAN — PART 3.B

3.B Agentic Systems Business Analysis**10 HOURS · 4 SESSIONS***Discovery, requirements, business case and engineering handoff for agentic systems.**Every session runs the same rhythm — concept framing, an instructor-led walkthrough, a guided hands-on lab, and a short review — within 2.5 hours.***SESSION 1** Day 5**2.5 hours****Objective** Map business workflows to agentic solutions and produce as-is / to-be process maps.

Topics Covered	Hands-On Lab (Applied)
▪ Mapping business workflows to multi-agent solutions – roles, handoffs, KPIs, decision points ▪ As-is to to-be process mapping – discovery interviews, swimlane diagrams, candidate automation identification	▪ Conduct three discovery interviews on a representative workflow; produce an as-is process map with timing, decision points and pain points
SESSION FLOW Concept framing & instructor walkthrough ~60 min · Guided hands-on lab ~75 min · Review, Q&A & wrap-up ~15 min	

SESSION 2 Day 6**2.5 hours****Objective** Model personas and author engineering-ready requirements with acceptance criteria.

Topics Covered	Hands-On Lab (Applied)
▪ Persona design and decision-point modelling – agent persona canvas, capability matrices, skill cards ▪ Requirements engineering for RAG, fine-tuning and tool-use – with explicit acceptance criteria	▪ Design a to-be agentic workflow with roles, KPIs, decision points, HITL escalation gates and a measured target cycle time ▪ Author engineering-ready requirements for RAG, tools and data with explicit acceptance criteria and edge cases
SESSION FLOW Concept framing & instructor walkthrough ~60 min · Guided hands-on lab ~75 min · Review, Q&A & wrap-up ~15 min	

SESSION 3 Day 7

2.5 hours

Objective *Frame governance and a defensible cost / risk / value business case.*

Topics Covered	Hands-On Lab (Applied)
▪ Responsible AI, governance and LLMOps considerations in agentic business cases ▪ Cost / risk / value framing – KPIs that survive leadership review, sensitivity analysis, ROI bounds	▪ Identify governance and Responsible-AI considerations; produce a risk-tier classification under the EU AI Act ▪ Build a business case with cost and value numbers; defend it in front of a peer reviewer
SESSION FLOW Concept framing & instructor walkthrough ~60 min · Guided hands-on lab ~75 min · Review, Q&A & wrap-up ~15 min	

SESSION 4 Day 8

2.5 hours

Objective *Apply feasibility heuristics and hand off BA artefacts cleanly to engineering.*

Topics Covered	Hands-On Lab (Applied)
▪ Feasibility heuristics – when an agent is the right answer vs simpler automation – when not to automate ▪ Handover discipline – making BA artefacts usable by engineering, design and governance teams	▪ Hand off the package to engineering; walk a peer reviewer through it and close every clarification raised
SESSION FLOW Concept framing & instructor walkthrough ~60 min · Guided hands-on lab ~75 min · Review, Q&A & wrap-up ~15 min	

04 SUGGESTED CAPSTONE PROJECTS

Ten suggested capstones spanning Insurance, Banking, Healthcare, Supply Chain and Logistics — each scoped for a group of up to three over 10–15 days on synthetic data. Every brief states the business problem, what to build, and the success metrics it will be judged on.

1 INSURANCE Claims Triage Agent

Business problem First-notice-of-loss intake and routing is manual and slow.

What to build LangGraph agent for intake, classification, routing and drafting with HITL gates and PostgreSQL checkpointing; MCP tools for a mock policy KB and CRM; LangSmith tracing.

Success metrics Routing accuracy, HITL-escalation correctness, end-to-end latency, recovery-from-failure rate.

2 INSURANCE Underwriting Assistant Agent

Business problem Underwriters spend hours gathering risk data and drafting notes.

What to build Agent that pulls risk data via MCP tools, scores and drafts underwriting notes; escalation routing and audit logs.

Success metrics Draft-quality (rubric), tool-call success rate, escalation correctness, token-budget adherence.

3 BANKING SME Loan Onboarding Agent

Business problem SME lending burns hours per application on eligibility, bureau pulls and memo drafting.

What to build Multi-step LangGraph agent with checkpointing, HITL gates and typed tools; custom MCP servers for bureau-score and credit-memo; failure recovery and idempotency.

Success metrics Task-automation rate, memo-quality (rubric), HITL precision, chaos-test resilience.

4 BANKING Dispute Resolution Agent

Business problem Card and payment disputes are gathered and drafted manually across systems.

What to build Agent integrating mock CRM / ticketing via MCP to gather, classify and draft dispute responses; escalation routing and audit trail.

Success metrics Classification accuracy, draft-quality, tool-integration success, simulated resolution time.

5 HEALTHCARE Prior-Authorization Agent

Business problem Prior-auth checks and drafting are slow and error-prone.

What to build Multi-agent flow (planner, rule-checker, drafter) checking coverage rules and drafting prior-auth requests with HITL and audit trails.

Success metrics Rule-check accuracy, draft completeness, HITL-escalation correctness, auditability.

6 HEALTHCARE Care-Pathway Recommendation Agent

Business problem Care-pathway suggestions must be governed and adoptable before engineering can build them.

What to build As-is / to-be maps, engineering-ready requirements and a governed LangGraph agent for pathway suggestions with HITL; risk-tier classification.

Success metrics Requirement completeness, governance sign-off coverage, recommendation accuracy, HITL correctness.

7 SUPPLY CHAIN Procurement Assistant Agent

Business problem Procurement steps (catalog checks, PO drafting, approvals) are manual.

What to build Agent integrating mock SAP Ariba / ServiceNow via MCP and MuleSoft to check catalogs, draft POs and route approvals; circuit breakers and idempotency.

Success metrics Tool-integration success, PO-draft accuracy, approval-routing correctness, resilience under outage.

8 SUPPLY CHAIN Supplier Risk Monitoring Agent

Business problem Supplier risk signals are scattered and reviewed reactively.

What to build Multi-agent system (collector, scorer, escalator) gathering synthetic signals, scoring risk and escalating; LangSmith observability.

Success metrics Risk-scoring accuracy, escalation precision / recall, trace completeness, token-budget adherence.

9 LOGISTICS Shipment Exception Resolution Agent

Business problem Exceptions need fast diagnosis against carrier rules and drafted resolutions.

What to build LangGraph agent diagnosing exceptions, checking carrier rules via MCP tools and drafting resolutions with HITL; failure recovery.

Success metrics Diagnosis accuracy, resolution-draft quality, HITL correctness, recovery rate.

10 LOGISTICS Freight Booking Coordination Agent

Business problem Quoting, booking and confirming across carrier systems is manual and error-prone.

What to build Multi-agent orchestration (quote, book, confirm) across mock carrier systems via MCP / A2A; idempotency, circuit breakers and escalation.

Success metrics Booking success rate, cross-system consistency, resilience under partial outage, end-to-end latency.

05 CAPSTONE GUIDELINES & SHARK TANK PRESENTATION

Each group takes one capstone to a defended Shark-Tank presentation. The capstone converts the specialization's learning into a demonstrated, review-tested system.

Team & Execution

- Groups of up to 3 participants.
- Timeline: 10–15 days from assignment to the Shark Tank defence.
- Use synthetic or public data only — no real customer data, PII or PHI. Generate realistic, real-format test data.
- Pick one domain per project — Insurance, Banking, Healthcare, Supply Chain or Logistics.
- The build must integrate and demonstrate the core competencies of this specialization.

What “done” means

- It runs — a working end-to-end system, not slideware or a notebook of fragments.
- It's measured — report real numbers against the project's success metrics (accuracy, latency, cost, quality).
- It's safe — at least one evaluation or safety component (guardrails, refusal, hallucination or drift check).
- It's reproducible — a code repository with a README that lets the panel run it.

Pre-Pitch Deliverables

- Code repository with a README — setup, run instructions and an architecture diagram.
- Evaluation report with the metrics named in the project brief.
- A working demo — live, or a recorded fallback in case of network failure.
- A 20-minute pitch deck and a one-page summary of architecture, trade-offs and limitations.

SHARK TANK PRESENTATION — 30 MINUTES

20 minutes presentation + 10 minutes Q&A. Present the working system and walk its architecture, trade-offs, evaluation evidence, cost and limitations. The panel challenges the team live; the team defends its decisions.

20-Minute Presentation — Suggested Split

Time	Segment
3 min	Problem & business case — the pain, who feels it, and the cost of the status quo.
4 min	Solution architecture — the pipeline / agents and which specialization competencies were integrated, and why.
7 min	Live demo — show it working on real-format (synthetic) inputs.
4 min	Results & evaluation — metrics against the baseline — quality, latency, cost.
2 min	Limitations & next steps — what it does not do yet, and what would come next.
10 min	Panel Q&A — the panel challenges architecture, trade-offs, evaluation and cost; every member defends their area.

Evaluation Rubric

Criterion	Weight	What is assessed
Functionality	25%	The system works end-to-end and does what it claims.

Criterion	Weight	What is assessed
Technical Depth	20%	Sound engineering and correct use of the specialization's tools and patterns.
Evaluation & Safety	15%	Rigorous evaluation, guardrails, and honest handling of limitations.
Usefulness	15%	Clear real-world applicability and business relevance.
Novelty	15%	Originality of the approach and problem framing.
Presentation	10%	Clarity of the pitch and quality of the defence under Q&A.